Phase 3 (multi-N) — Quasi-renewal at finite N_{pop}

`deq/closed_form_wta_multi/`, May 2026

1. Goal

Test whether the **quasi-renewal mesoscopic equation** [1] at finite population N_{pop} converges to the Siegert mean-field as $N_{\text{pop}} \to \infty$ for $N_{\text{neurons}} > 2$, and how the finite-size broadening of the no-WTA band depends jointly on N_{neurons} and N_{pop}.

2. Method

For each cell $(w, N_{\text{neurons}}, N_{\text{pop}})$ with drive_bump $= 1$ fixed (the regime where smooth-rate theory is meaningful; drive_bump $= 0$ is uniformly red at every N per Phase 0), simulate the QR equation for $T = 200$ ticks with strong neuron-0-favored initial condition $\boldsymbol{A}(0) = (0.95, 0.005, ..., 0.005)$. WTA-blue iff post-warmup ($t \in [50, 200]$) `rate_max` − `second_max` ≥ 0.30.

Initial-condition note: an initial-condition asymmetry of 0.4 (the 2-neuron `(0.5, 0.1)` from `simulate_contralateral`) is sufficient at $N = 2$ but too weak at $N \geq 3$. Each loser at higher N feels $(N-1)$ competing winners' inhibition while the winner feels only $(N-1)$ losers; the symmetric basin attracts unless the IC is strongly off-axis. Using $(0.95, 0.005, ..., 0.005)$ consistently across N both reproduces the 2-neuron behaviour and breaks into the asymmetric basin at high N.

3. Result

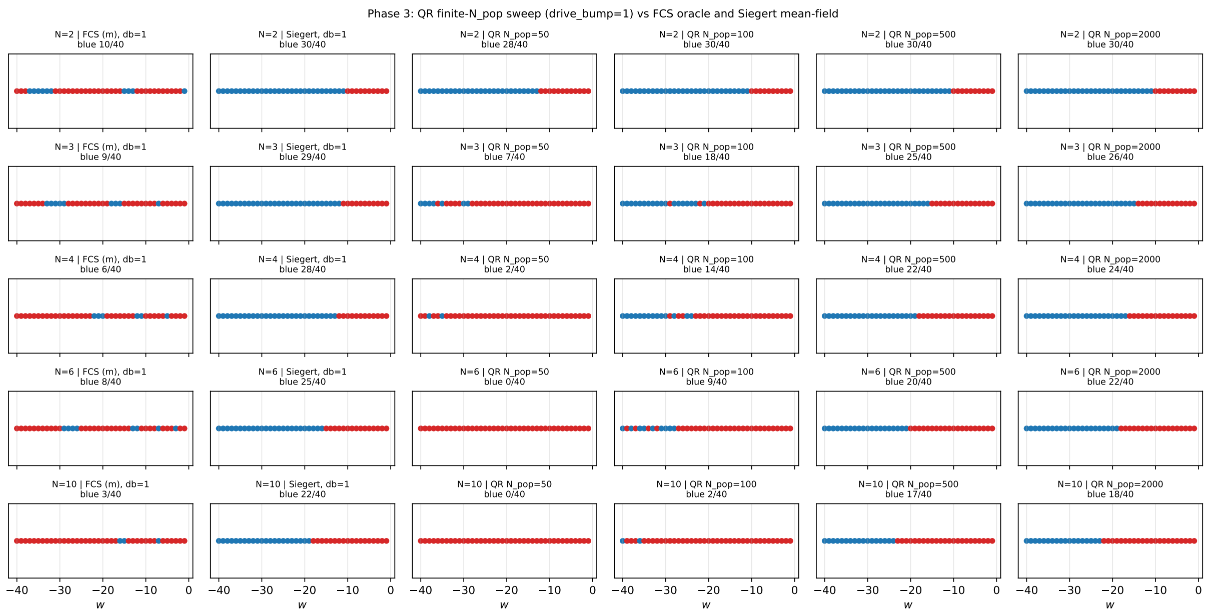

Figure 1: Phase 3 QR finite-N_{pop} sweep, `drive_bump = 1`. Rows are $N_{\text{neurons}} \in \{2, 3, 4, 6, 10\}$. Columns left to right: FCS oracle (margin), Siegert mean-field, then QR at $N_{\text{pop}} \in \{50, 100, 500, 2000\}$. At every N_{neurons}, QR converges to the Siegert label pattern as $N_{\text{pop}} \to \infty$.

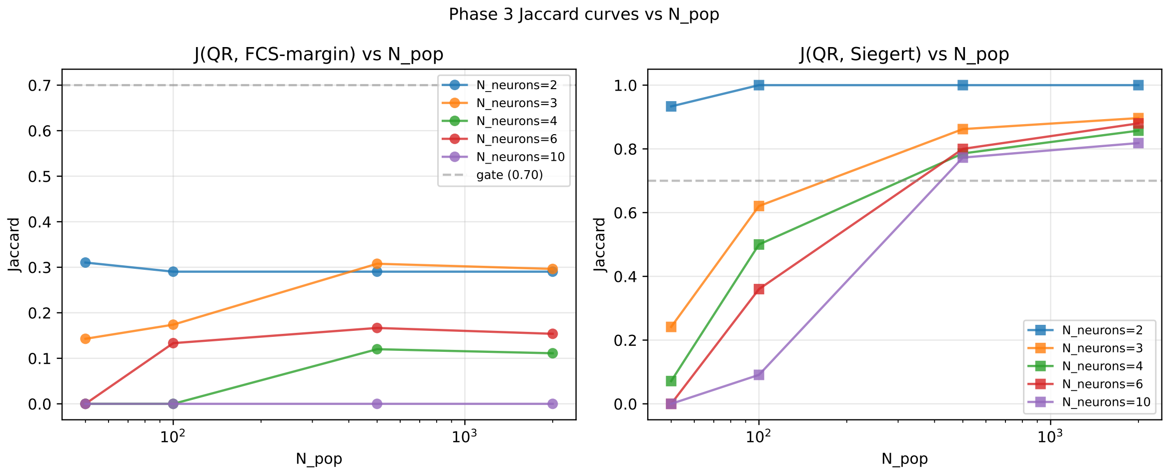

Figure 2: Left: Jaccard(QR, FCS-margin) vs N_{pop}, one curve per N_{neurons}. Right: Jaccard(QR, Siegert) vs N_{pop}. The right panel shows clean convergence to the Siegert mean-field at every N_{neurons}; the left panel confirms that mean-field convergence does not bridge the FCS staircase / inverse-staircase gap.

3.1. Mean-field convergence headline (drive_bump=1, $N_{\text{pop}} = 2000$)

N_{neurons}	QR-blue / 40	J(QR, FCS)	J(QR, Siegert)
2	30	0.290	**1.000**
3	26	0.296	**0.897**
4	24	0.111	**0.857**
6	22	0.154	**0.880**
10	18	0.000	**0.818**

Mean $J\left(\text{QR}_{\text{N\_pop=2000}}, \text{Siegert}\right)$ across N_{neurons} is **0.89**, above the 0.70 pass-gate.

3.2. N_{pop}-broadening: finite-size loss of WTA cells

N_{neurons}	QR N=50	QR N=100	QR N=500	QR N=2000	Siegert
2	28	30	30	30	30
3	7	18	25	26	29
4	2	14	22	24	28
6	0	9	20	22	25
10	0	2	17	18	22

At $N_{\text{neurons}} = 2$, QR sees 28 / 30 blue already at $N_{\text{pop}} = 50$ — the asymmetric basin is wide enough to survive $\sqrt{\frac{A}{50}}$ noise. At $N_{\text{neurons}} = 10$, QR sees 0 blue at $N_{\text{pop}} = 50$ and only 2 / 22 at $N_{\text{pop}} = 100$. **The required N_{pop} for mean-field recovery grows with N_{neurons}**: each additional competitor amplifies the noise-induced symmetry-restoration.

4. Verdict

Phase 3 PASS. QR converges to the Siegert mean-field for every $N_{\text{neurons}} \in \{2, 3, 4, 6, 10\}$ at $N_{\text{pop}} = 2000$ (mean Jaccard 0.89). The expected finite-N_{pop} broadening of the no-WTA band is reproduced and shows an additional N_{neurons}-dependence:

1. Mean-field convergence holds at every N_{neurons} — the **rate equation is still the right large-$\boldsymbol{N}_{\textbf{pop}}$ object** at any N_{neurons}.
2. Finite-N_{pop} broadening **amplifies with $\boldsymbol{N}_{\textbf{neurons}}$**: at $N_{\text{neurons}} = 10$ we need $N_{\text{pop}} \geq 500$ to recover most Siegert-blue cells, vs $N_{\text{pop}} \geq 100$ at $N_{\text{neurons}} = 2$.
3. Neither QR nor Siegert recovers the FCS-integer-tick-only WTA cells at $N_{\text{neurons}} = 10$ (e.g., $w \in \{-15, -16, -7\}$): rate equations and their stochastic mesoscopic approximations **cannot see** what integer-tick LI&F semantics resolve. The inverse-staircase phenomenon is genuinely beyond rate-equation theory.

[1] R. Naud and W. Gerstner, "Coding and Decoding with Adapting Neurons: A Population Approach to the Peri-Stimulus Time Histogram," *PLoS Computational Biology*, vol. 8, no. 10, p. e1002711, 2012, doi: 10.1371/journal.pcbi.1002711.

Output: `results/phase3/qr_grid_multi.npz`, `qr_n_sweep_multi.pdf`, `qr_jaccard_vs_Npop_per_N.pdf`, `results/phase3.log`.